CIS 613: System Testing

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System Testing

- "Intuitively clear"
  - customer expectations
  - close to customer acceptance testing
- BUT we need a better basis for really understanding system testing
- Threads—the subject of system testing
- How are they identified?
  - ad hoc?
  - from experience?
  - from a possibly incomplete requirements specification?
  - from an executable model? (Model-Based Testing)

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Threads of Execution

- An execution time concept
- Per the definition, a thread can be understood as a sequence of atomic system functions.
- When a system test case executes
  - a thread occurs, and
  - can be observed at the port boundary of the system
- The BIG Question: where do we find (or how do we identify) threads?
- Our approach—Model-Based Testing

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Views of Threads

- A scenario of normal usage
- A use case or user story
- A stimulus/response pair
- Behavior that results from a sequence of system-level inputs
- An interleaved sequence of port input and output events
- A sequence of transitions in a state machine description of the system
- An interleaved sequence of object messages and method executions
- A sequence of machine instructions
- A sequence of source instructions
- A sequence of MM-paths
- A sequence of atomic system functions (to be defined)

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Sources of Threads

- A “Expanded Essential Use Case”
  - pre-conditions
  - interleaved sequence of input and output events
  - post-conditions
- A path in an executable model
  - Finite State Machine
  - Event-Driven Petri Net

We are going to look at paths in a FSM

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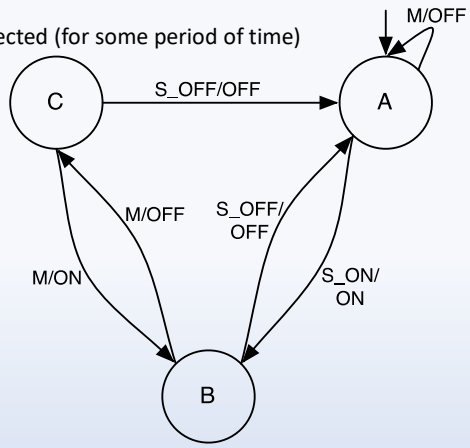
What are the threads here?

Input events: (M, S_ON, S_OFF)

- M = Motion Detected / Not Detected (for some period of time)
- S_ON = Switch Turned On
- S_OFF = Switch Turned Off

Output events (ON, OFF):

- Lights Off
- Lights On



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Always the Case

When there is not Enough Time for Testing

- Operational Profiles
  - Need “traffic” estimates
  - Focus testing on the most likely modes of system usage
- Risk-Based Testing
  - An extension to Operational Profiles
  - Still need “traffic” estimates
  - Also need cost penalties for failed transactions
  - Focus on testing “the most potentially expensive” failures

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Operational Profiles

- Zipf’s Law: 80% of the activities occur in 20% of the space
  - productions of a language syntax
  - natural language vocabulary
  - menu options of a commercial software package
  - area of an office desktop
  - floating point divide on the Pentium chip
- Rationale for operational profile testing
  - a small fraction of all possible threads represents the majority of system execution time
  - find the occurrence probabilities of threads and use these to order thread testing.

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Operational Profiles

- A shortcut when test time is limited
- Estimate transition probabilities in a high level finite state machine
- Product of probabilities on a path is the probability of that path executing
- Method
  - Estimate probabilities
  - Compute path probabilities
  - Sort paths most to least likely
  - Test most likely paths until time expires

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FSA Path Probabilities

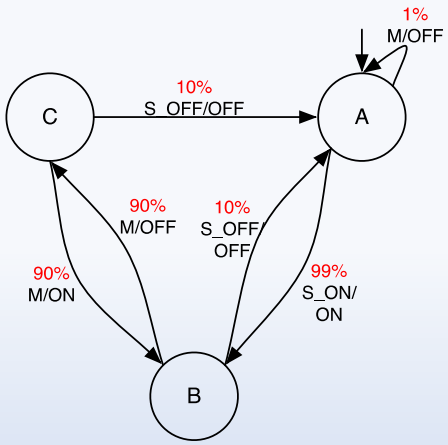
Automatic Light Switch Finite State Machine

Thread (State, Action):

- A, Turn switch on (S_ON)
- B, Leave room (M)
- C, Enter room (M)
- B.

Probability (State, Action = %):

- A, S_ON: 99%
- B, M: 90%
- C, M: 90%
- B = 99%*90%*90% = ~80%



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In Class Assignment: Identify Thread Chances

Automatic Light Switch Finite State Machine

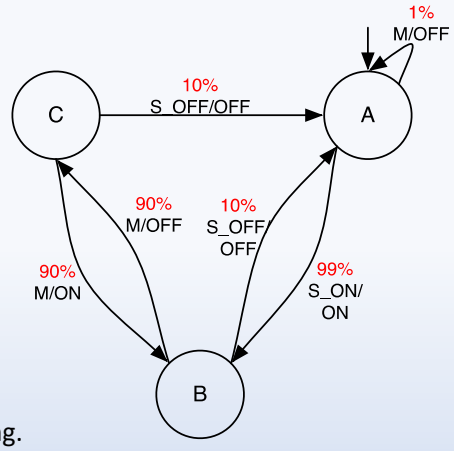
Identify 3 threads:

1. One should be *likely*
2. One should be *unlikely*
3. One should be *between*

For each thread, document the:

- States,
- Transitions, and
- Likelihood.

Threads should be ***four*** states long.



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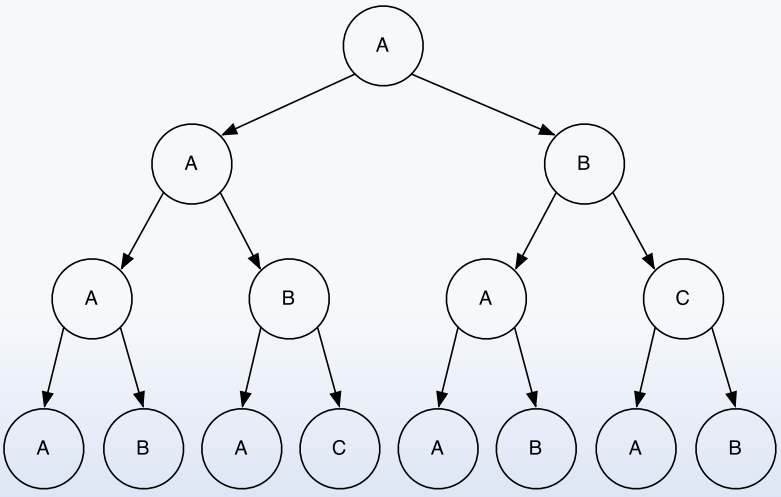
In Class Assignment: Identify Thread Chances



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Risks in Software

- A risk is an unwanted event that has negative consequences (often due to incorrect estimation)
- Distinguish risks from other project events
  - **Risk impact**: the loss associated with the event
  - **Risk probability**: the likelihood that the event will occur
- Quantify the effect of risks
  - **Risk exposure** = (risk probability) x (risk impact)

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Risk-Based Testing

Probability \ Impact	Negligible	Marginal	Critical	Catastrophic
Certain	High	High	Extreme	Extreme
Likely	Moderate	High	High	Extreme
Possible	Low	Moderate	High	Extreme
Unlikely	Low	Low	Moderate	Extreme
Rare	Low	Low	Moderate	High

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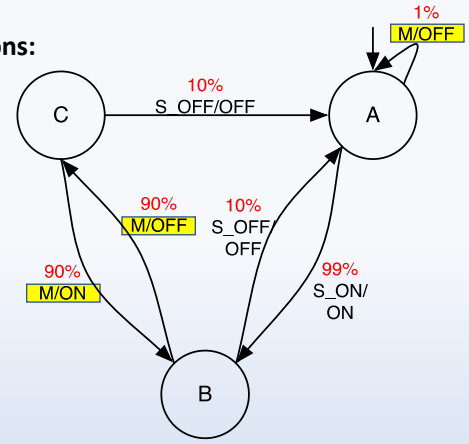
FSA Path Probabilities + Risks

Automatic Light Switch Finite State Machine

Risky (Motion-Based) Transitions:

- A to A,
- B to C,
- C to B



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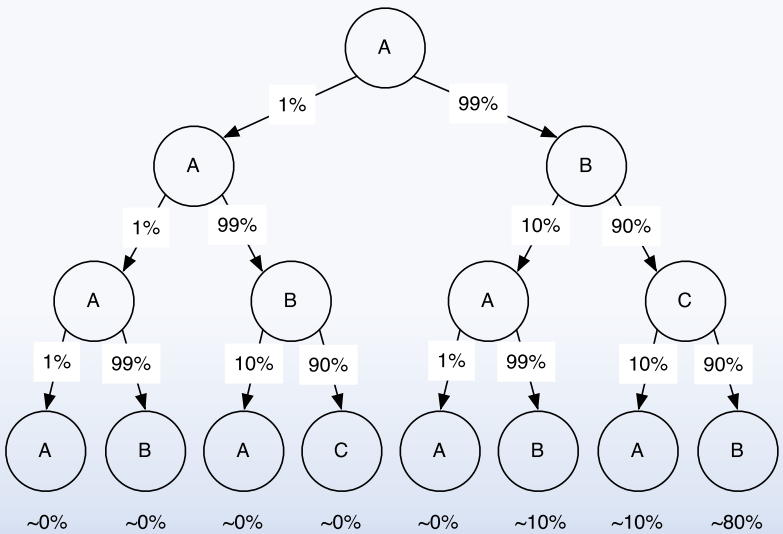
Given the likelihood, add the risk!



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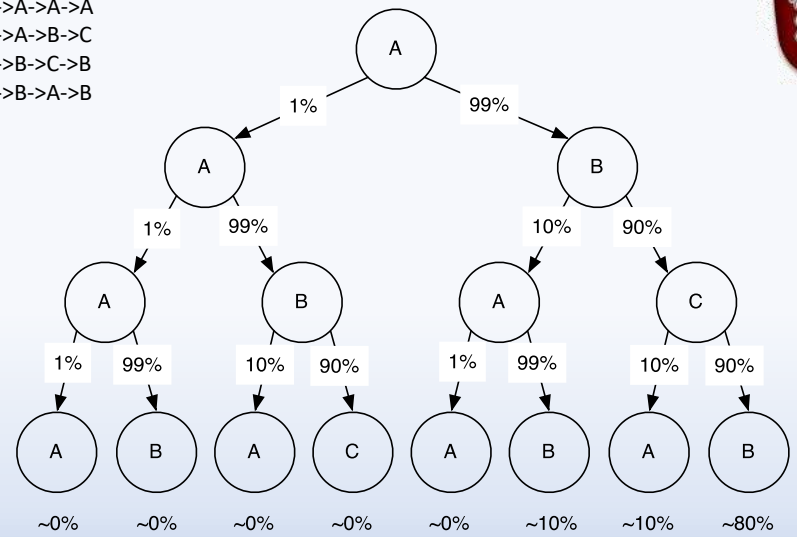
What path has the most risk probability?

A. A->A->A->A
B. A->A->B->C
C. A->B->C->B
D. A->B->A->B



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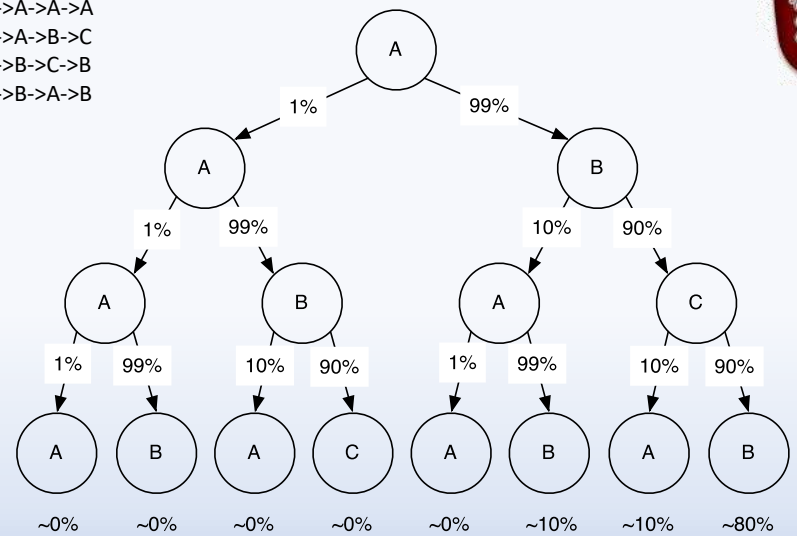
What path has the most risk exposure?

A. A->A->A->A
B. A->A->B->C
C. A->B->C->B
D. A->B->A->B



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Discussion

- How does system testing differ from unit / module testing?
- How does system testing differ from integration testing?
- Could tests cover both unit, integration, and system testing?

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